Problem Set 8

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Hopfield model

In the Hopfield network, we make even a more drastic assumption that the activity of a neuron is a binary variable, $s_i \in [+1, -1]$. In the deterministic version of this model, the state of a cell is set according to the following equation

$$s_i(t + \Delta t) = \mathbf{sgn}(\sum_j w_{ij} s_j), \tag{1}$$

We consider an asynchronous update rule for each neuron (choosing either a random or fixed update order and update s_i according to that order at each clock cycle).

(1) Check my lecture note and proof that any mixture of an odd number of memory patterns, such as

$$\xi^{mix} = \mathbf{sgn}(\xi^{\mu_1} + \xi^{\mu_2} + \xi^{\mu_3}) \tag{2}$$

are fixed point of the Hopfield model.

(2) Numerically simulate the dynamics of Hopfield network with random unbiased memory patterns, using 500 binary neurons. We want to draw some statistical conclusions, so in the tasks that follow, you should run many simulations using different sets of random stored patterns and, for each, start from many different initial conditions (so I recommend writing reasonably efficient routines, e.g. minimize the number of for-loops in MATLAB). For simplicity, you can choose the order of update of the neurons by going through neuron 1 to N all the time.

i) Probability of perfect recall: Store 10 patterns, run simulations to find the probability of perfect recall. That is, start with a pattern, corrupt some fraction of its bits, input the corrupted pattern to the network, and wait until the network settles into a fixed point. Plot the fraction of times you get perfect recall of the original state against the fraction of corrupted bits.

ii) Memory retrieval with errors: Start the dynamics with one of the memory states. Plot the fraction of errors (fraction of incorrectly recalled bits) in the final state as a function of the number of stored patterns P. Change P from 10 to 100.

E-I balanced firing rate model

Consider a large recurrent neural network with two populations E and I with identical number of neurons N. Let us assume that each neuron will randomly send K outputs to the E population and K outputs to the I population. The synaptic weights are given by J^{EE}, J^{II}, J^{IE}, J^{EI}. We consider the strong weight limit, namely $J^{AB} = w^{AB}/\sqrt{K}$, where A, and B indicates either the E population or the I population, and w^{AB} is an O(1) number independent of the number of connections.

In addition to recurrent connections, each neuron also receives constant external synaptic input $h_0^A = \sqrt{K} w_0^A$, where w_0^A is an O(1) number that can be different for E population and I population. The evolution of the neural activity is governed by the following equations as discussed in the lectures:

$$\begin{aligned} \tau \frac{dI_i^E}{dt} &= -I_i^E + \sum_{j=1}^{N} J_{ij}^{EE} r_j^E(t) - \sum_{j=1}^{N} J_{ij}^{EI} r_j^I(t) + h_0 \\ r_i^E(t) &= [I_i^E(t))]_+ \end{aligned} \tag{3}$$

where $\tau = 10$ ms, and we use a Relu like activation function. We consider the diluted recurrent neural network, namely $K \ll N$. For example, consider $K = pN$, where $p = 0.1$. A similar equation can be written for the I population.

(1) Define the following parameters: such that $w^{EI} = \alpha_E w^{EE}$, and $w^{II} = \alpha_I w^{IE}$, $w_0^E = \beta_E w^{EE}$, $w_0^I = \beta_I w^{IE}$. Look at my slides and figure out under what parameter regime the system has stable dynamics, and confirm this with computer simulation.

(2) Increase the external input, say $h_0 = \sqrt{K} w_0^A m_0$, and increase m_0 in a reasonable range between 0 and 1, how does the average neural activity in E and I activity changes.

(3) Compute the autocorrelation function of the neuron's own activity over time $C(\tau) = \langle r_i(t) r_i(t+\tau) \rangle$ and pairwise covariance between neurons $\langle r_i(t) r_j(t) \rangle - \langle r_i(t) \rangle \langle r_j(t) \rangle$. If you have time, please compute the distribution of covariance as α_E approaches 1.